

QUESTIONS AND ANSWERS



Red cherry fruits

Q Why are the fruits of the cherry plant red?

A The fruits of the cherry plant are bright red in color to attract birds to eat the fruits. The cherry fruits contain seeds that have a hard protective covering. This ensures that when the seeds are eaten by birds they pass unharmed through the digestive system of the creatures. In this way, the seeds are safely spread, and the plant guarantees the survival of its offspring. Plants pollinated by insects are rarely red because insects, with the exception of butterflies, cannot see the color red.

Q Which plant is considered to be the most bizarre of all?

A *Welwitschia mirabilis*, also known as the tumboa, from the Namib Desert in Africa, is one of the strangest plants in the world. Known to live for up to 2,000 years, it has a stumpy stem and just two straplike leaves, which grow nonstop throughout its life. As the plant ages, its leaves become twisted and gnarled, and they eventually can be many yards long. The leaves are tough and woody—an adaptation that helps to stop them from being eaten, or drying out. *Welwitschia* survives in a region where there is little rain, but where fog rolls in from the sea. The plant's leaves gather moisture from the fog, helping it to survive. *Welwitschia* does not grow flowers, but produces seeds in cones.

Q What is the richest plant region of the world?

A South America, which holds an estimated 90,000 species, is the world's richest plant region. Brazil is the country with the greatest known number of plant species, at 56,000, followed by Colombia, with 35,000 species. Mexico, Venezuela, Ecuador, Bolivia, and Peru are not far behind. The proliferation of plant species in this part of the world is thought to be due to the moist habitat associated with its tropical rainforests, as well as the relatively recent arrival of humans.

Q Why do some tree leaves change color in autumn?

A As the days become colder and shorter, chlorophyll, the green pigment in the leaves, breaks down and flows back into the tree. Meanwhile, waste products, such as tannins, pass out into the leaves. This chemical change produces browns and reds in the colors of the leaves as they die. Trees that lose their leaves are said to be deciduous.

Record Breakers

Smallest plant

- The world's smallest flowering plant is duckweed, (*Wolffia angusta*). A tablespoon can hold more than 100,000 plants, with each measuring only $\frac{1}{30}$ in (0.8 mm) long and $\frac{1}{60}$ in (0.4 mm) wide.

Largest seed

- The largest seed produced by any plant is that of the coco-de-mer (*Lodoicea maldivica*), from the Seychelles. This palm, also known as the double coconut tree, produces seeds that weigh up to 50 lb (23 kg) and take up to 10 years to grow into a tree.

Tallest tree

- The Mendocino coast redwood (*Sequoia sempervirens*) found in California, is the world's tallest tree, reaching a maximum height of 367½ ft (112.01 m).



Duckweed



Welwitschia mirabilis